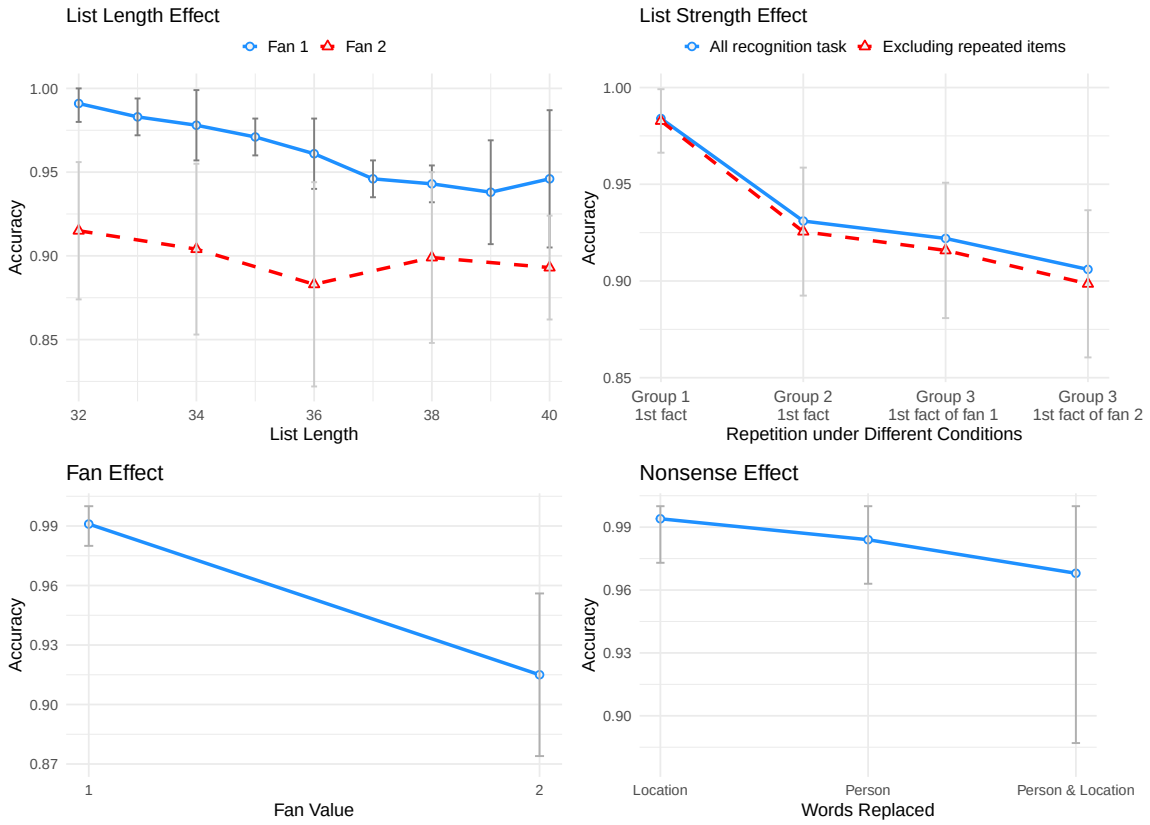

Figure 1: Performance of the online language model (‘`gpt-4-0125-preview`’) on Dataset 1. List length effect: Study lists varied in size (32~40 person–location pairs), showing that recall accuracy declines as list size increases. List strength effect: Repetition of selected study facts was implemented, revealing that the accuracy of weaker items is consistently lower than that of stronger items, yet should not be affected by their presence. Fan effect: Fixing the study list size at 32 pairs and varying fan values demonstrated that higher associations to an item reduce recall accuracy. Nonsense effect: Replacing person and/or location words with nonsensical tokens showed minimal impact on recognition accuracy, suggesting robustness of model performance to meaningless inputs.

in humans. The recall proportion of critical lure words is 0.114, suggesting that language models are likely to generalize these lure words based on the associations in the study list, even though the lure words are not presented in the study list.

In the position effect experiment, we observed that language models can accurately recall all the words from the study list with perfect recall accuracy (1), regardless of changes in the order of the words within the study list. In contrast, humans exhibit primacy and recency effects during recall tasks (see Figure 3c). Participants tend to remember the words presented at the beginning and the end of the list more clearly, while recall accuracy for words in positions 4 to 8 is relatively lower in Roediger’s experiment [38].

Comparison Analysis with Offline Models

While much recent research focuses on online LLMs accessible via APIs, such as GPT-4, an important complementary direction involves understanding the capabilities of offline LLMs, which can be deployed locally without reliance on external servers. Unlike online models, which are often optimized for interaction, offline models provide greater reproducibility and control over the inference process. In the ablation study, we extend our investigation of memory effects by applying frameworks from human memory research to offline LLMs, aiming to evaluate whether similar cognitive patterns exist under constrained conditions. To enhance the possibility that these models can fully engage with

Table 1: Recognition Results for Studied Items and Critical Lures in Dataset 2

Items types and condition	Proportion of Old response
	Overall
study + recall	1.000
study + arithmetic	1.000
non-studied	0.053
critical lure	0.114

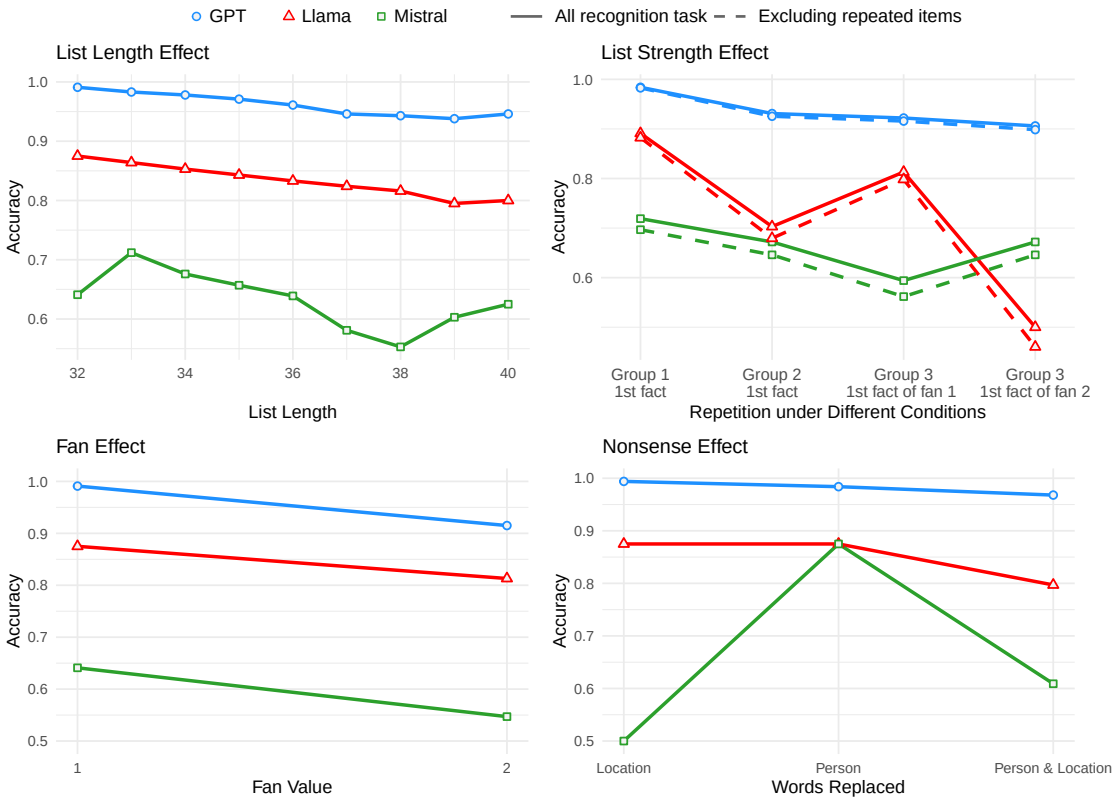

Figure 2: Performance comparison between online and offline LLMs on Dataset 1. Offline models (both Mistral and Llama) exhibit similar patterns across the four effects, despite some fluctuations for Mistral under the list length effect and for Llama under the list strength effect. However, online LLMs consistently outperform offline models, demonstrating more robust and reliable performance across conditions. In addition, due to the frozen parameters of offline models, repeated trials with identical input prompts do not change the output. As a result, Mistral may not perform well in some particular tasks of the nonsense effect.

our memory tasks, we selected state-of-the-art models with relatively large parameter counts. Specifically, we experimented with `'Mistral-7B-Instruct-v0.3'`, and `'LLaMA-3-8B'`. Across all seven experiments spanning two datasets, we obtained complete results using each of these three offline models. A subset of the comparison results is presented in Figure 2.

We observed that for the four effects in Dataset 1—list length/strength, fan effect, and nonsense effect—the offline models exhibited a similar trend, despite some fluctuations (e.g., Mistral under the list length effect and Llama under the list strength effect). Nevertheless, online LLMs consistently outperformed offline LLMs. Additionally, because the parameters of offline models are frozen and can not be further fine-tuned, repeated trials with identical input prompts do not change the outputs. Hence, Mistral shows limited performance in certain tasks under the nonsense effect, such as when only locations are replaced or when both persons and locations are substituted. For the following effects in Dataset 2, the offline models demonstrated perfect performance on the position effect as well, accurately recalling and recognizing words at all positions from the associated word lists. As for the DRM effect, offline models achieved a hit rate of 1, which is similar to GPT. Nevertheless, the offline models exhibited poor generalization ability, as they consistently recognized lure words as new rather than previously encountered items, with the probability of identifying lure words as old words being effectively zero. This suggests that offline models were unable to generalize lure words based on the associated words they studied.

Comparison of LLM and Human Memory Effects

In this work, we use human memory research as an experimental toolkit to gain insights into the functioning of language models. By aligning established effects of human recall and recognition with LLM behavior, we aim to better understand the extent to which these systems exhibit similar memory patterns. By drawing on canonical experimental paradigms from cognitive psychology, we also compare empirical human data examples with model behavior. Rubin *et al.* showed in Figure 3a that human memory decays in a predictable way over time, with recall probability decreasing sharply within the first minute post-encoding before stabilizing [40].

Table 2: Comparison of Human and LLM Performance on Key Memory Effects

Memory Effect	Human Behavior	LLM Behavior
List Length Effect	Recall accuracy decreases as list length increases.	✓ Shows a parallel decrease in overall performance.
List Strength Effect	Repetition of certain items leaves recognition of other items unaffected.	✓ Shows a parallel pattern, with weaker items' recognition remaining unaffected.
Fan Effect	Retrieval accuracy decreases with the number of associations.	✓ Mimics the fan effect in memory tasks, showing less accurate outputs with increased fan.
Nonsense Effect	Meaningless items are harder to recall.	✗ Superior robustness to semantically meaningless content.
Positional Effect	Primacy, and recency effects are observed.	✗ Position-invariant, indicating a divergence from human-like serial position effects.
DRM-style False Memories	Tends to falsely recall related but non-studied items.	✓ Exhibits lure responses consistent with the DRM paradigm, indicating similarity.
Cross-domain Generalization	Flexible generalization with abstraction.	✓ Exhibits generalization but sometimes overfits specific examples, partially consistent.

For instance, according to the list length effect, recall accuracy declines with increasing list length. There is a comparable declining trend in both human and model performance, and this alignment indicates that similar memory interference mechanisms exist inside LLMs despite the structural differences between LLMs and the human brain. Furthermore, the list strength effect is not manifested in LLM behavior, as the accuracy of weaker items is not influenced by the repeated items, which is consistent with human behavior in recognition tasks.

As demonstrated in the associative recognition task modeled by the ACT-R (Adaptive Control of Thought-Rational) and SEF (Shifted Exponential Function) frameworks, the fan effect also appears in the trade-offs between accuracy and reaction time in human cognition [46]. As shown in Figure 3b, the accuracy for the Fan 1 group consistently surpasses Fan 2, especially at longer lags, illustrating the stabilizing effect of reduced interference and longer reaction time. These results are supported by our experiments: When there is less association interference, LLMs demonstrate improved discriminability in recall tasks, leading to better performance. This suggests that LLMs may encode information with associative spread similar to human models of memory, further validating their utility as cognitive models.

A key divergence arises in sensitivity to information positional order. Generally, as shown in Figure 3c, human memory shows a U-shaped serial position curve [21], with higher recall at the beginning and end of lists [38]. In contrast, LLMs show no sensitivity to the positional bias.

Overall, our findings, as summarized in Table 2, reveal that language models exhibit human-like patterns in phenomena such as the fan effect, list length/strength effect, and DRM effect. However, their performance appears to be less influenced by factors such as nonsense effects and positional effects. Additionally, large language models also exhibit human-similar generalization ability, as they show a high proportion of generalizing critical lure words even when these words are not presented in the study list. An interesting result is that language models tend to exhibit memory errors similar to humans in recognition tasks, yet their performance in recall tasks remains nearly perfect. The observed superior performance of LLMs across part of the tested memory effects may originate from their underlying cache mechanism, an architectural feature that functionally parallels the 'tape recorder' in humans, where external memory augmentation enhances recognition.

Comparison of LLM and Human *d'* Values

A core quantitative measure of recognition memory sensitivity is *d'*, derived from signal detection theory [**?**]. It quantifies how well a participant/system can discriminate between *targets* (studied items) and *foils* (unstudied items), independent of response bias. The formula of *d'* is shown in Equation 1 for our LLM experiments,

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F), \tag{1}$$

where H and F denote the hit and false-alarm rates (in Equation 4 and 3), respectively, and Φ^{-1} is the inverse of the cumulative normal distribution. A higher d' value indicates superior discriminability—meaning the system reliably distinguishes studied from unstudied pairs—while a d' close to zero reflects near-chance performance and weak memory sensitivity.

In classic human associative-recognition studies such as Schneider and Anderson [46], d' values typically range from 2.0–4.0, depending on fan size and retrieval delay: the mean accuracy for Fan 1 conditions converges toward 3.7, while Fan 2 conditions reach a stable value near 2.5, indicating the impact of increased associative interference. as shown in Figure 3b. Meantime, our LLM experiments (in Figure 4) yielded comparable patterns: online language model achieved mean d' values of approximately 4.1 for Fan 1 and 3.1 for Fan 2, exhibiting human-like discriminability patterns. This alignment suggests that LLMs encode associative dependencies in a way similar to human interference dynamics, consistent with cognitive-model performance. Additionally, the *d'* values observed in online LLM were higher than those reported in human studies, indicating that LLMs exhibit stronger memory sensitivity.

The first subfigure of Figure 4 illustrates the list length effect, showing a decline in *d'* value as the number of studied pairs increases. This downward trend aligns with human findings on the list length effect. Fan 1 consistently outperforms Fan 2 across all list lengths, indicating that less association impose less retrieval